

Project Workflow: Week 1

Phase: Project Planning & Concept Finalization

1. Objective

The primary goal of Week 1 was to establish a robust foundation for a feature-rich, 2D top-down car racing game built entirely in Java using Swing and AWT graphics. This included:

- Defining the project scope, core gameplay loop, and win/lose conditions.
- Selecting an optimal technology stack for smooth real-time rendering and physics.
- Conceptualizing the road system, environment design, and overall visual identity of the game.

2. Detailed Task Breakdown

Feature Identification

Conducted research on top-down arcade racing games to finalize the core feature set: a 3-lane scrolling road with curve physics, player lane-switching with speed boosting, randomized AI traffic, gas pickups, a scoring system, and a 5,000-unit distance win condition.

Technical Stack Selection

Selected Java SE with Swing and AWT graphics for all rendering, requiring no external libraries or game engine. A `javax.swing.Timer` firing at ~60 FPS was chosen to drive the core update/repaint game loop.

UI/UX & Environment Design

Sketched the 900x700 game window layout, the 3-lane road with animated lane markings, and a parallax-scrolling environment featuring trees, bushes, mountains, clouds, lakes, and parks. Planned a dynamic sky that transitions from dawn to afternoon to dusk as the player progresses, along with a HUD overlay for score, speed, and distance.

Architectural Setup

Designed a modular, object-oriented class architecture: a JFrame entry point, a GamePanel for the core loop and rendering pipeline, an Environment manager composed of scenery classes (Bird, Cloud, Mountain, Lake, Park, Tree, Bush), a Road class for curve physics, an abstract Car class extended by PlayerCar and EnemyCar, and a GasItem class for collectible pickups.

3. Deliverables

- Project Scope Document: A formal outline of the game's features and limitations.
- UI Wireframe Sketches: Conceptual layouts for the road, HUD, and environment scenery.
- Class Architecture Diagram: A finalized object-oriented structure for the game's core classes.
- Project Timeline: A structured 10-week development roadmap.

4. Tools & Resources

Development Environment

- Java JDK 8+
- IntelliJ IDEA / VS Code

Reference Materials

Java Swing, AWT 2D graphics, and javax.swing.Timer-based game loop references.